

DAILY UPDATE: MATH 101: LINEAR & MULTILINEAR ALGEBRA (FALL 2026)

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1. WEEK 1

1.1. Monday, September 14, 2026. Our first class! This course is a modern introduction to the study of linear and multilinear algebra. A broad theme throughout this class is that linear algebra over a field then appears as one especially nice case of a more general story about rings and modules. We worked on Worksheet 1.1, which introduces rings, ring homomorphisms, and ideals. We began by deriving basic consequences of the ring axioms, such as the uniqueness of identities and inverses, the identity $a0 = 0$, and the fact that the units $R^\times$ form a group, and then looked at first examples including $\mathbb{Z}/n\mathbb{Z}$, polynomial rings $\mathbb{K}[x]$, and matrix rings $\text{mat}_n(\mathbb{K})$, paying attention to units, zero divisors, and noncommutativity (e.g., $E_{12}E_{21} \neq E_{21}E_{12}$). We then defined ring homomorphisms together with their kernels and images, and observed that kernels are two-sided ideals. This led to the construction of the quotient ring R/I and the First Isomorphism Theorem for rings. We needed by consider a few more nice examples.

1.2. Wednesday, September 16, 2026. We worked on Worksheet 1.2, which introduces modules, the central objects of this course. A module over a ring R is an abelian group with a compatible action of R . This generalizes a vector space by allowing the scalars to come from any ring. We saw that $\mathbb{K}$ -modules are exactly $\mathbb{K}$ -vector spaces, that every abelian group has a unique $\mathbb{Z}$ -module structure, and that R and its ideals are R -modules. We defined submodules, R -linear maps, kernels, and images, etc. We looked at a particularly important example in detail, namely that we may think of a vector space V and linear map T as allowing us to put a $\mathbb{K}[x]$ -module structure on V . We then constructed quotient modules M/N and established the First Isomorphism Theorem. There are three more isomorphism

theorems, the Second and Third Isomorphism Theorems and the Correspondence Theorem, be sure to read about them. *Expect a quiz on Monday.*

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